

DATA ANALYTICS & BI CAPSTONE

Sales Performance Dashboard

DOMAIN

Data Intelligence

Python & Pandas

PROJECT OBJECTIVES



Sales & Revenue

Analyze high-level sales and revenue drivers automatically.



Top Categories

Identify top products and regional growth performers.



Time Trends

Evaluate sales seasonality across different quarters.

PROBLEM STATEMENT & SOLUTION



Challenges



Difficult manual processing of large sales files.



Lack of centralized KPI monitoring.



Time-consuming manual report building.



Proposed Solution

An automated **Sales Performance Dashboard** converting raw transaction datasets into:

- Interactive visual reports
- Instantaneous KPI calculations
- Instant trend discovery

DATASET STRUCTURE

Data Attributes

Order ID & Date

Product Category

Sales Region

Quantity Sold

Revenue (\$)

Profit Margin (\$)

- Structured transaction log containing regional sales performance.
- Supports instant calculations for executive KPIs and trend discovery.

A	B	C	D	E	
	▼ Date ▼↑	Regi▼	City	▼ Categor▼	Pro
007351	1-Jan	East	Boston	Bars	Can
007352	4-Jan	East	Boston	Crackers	Wh
007353	7-Jan	West	Los Angeles	Cookies	Cho
007354	10-Jan	East	New York	Cookies	Cho
007355	13-Jan	East	Boston	Cookies	Arro
007356	16-Jan	East	Boston	Bars	Can
007357	19-Jan	East	Boston	Crackers	Wh
007358	22-Jan	West	Los Angeles	Bars	Can
007359	25-Jan	East	New York	Bars	Can
007360	28-Jan	East	New York	Snacks	Bot

ANALYTICAL LIFECYCLE



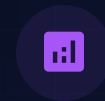
1. Ingestion & Clean

Removal of duplicates & missing values.



2. Transformation

Formatting dates and generating KPIs.



3. Visual Dashboard

Interactive aggregation for executive insights.

DATA PROCESSING ENGINE

pipeline_engine.py

```
# 1. Load Raw Dataset

import pandas as pd
df = pd.read_csv("sales_performance_data.csv")

# 2. Data Cleaning & Normalization

df.drop_duplicates(inplace=True)
df['Order Date'] = pd.to_datetime(df['Order Date'])

# 3. Feature Engineering & KPIs

df['Profit_Margin'] = (df['Profit'] / df['Sales']) * 100

# 4. Aggregations & Outputs

summary = df.groupby('Region').agg({'Sales': 'sum'})
```

Execution Logic



Data SanitizationCleans duplicates and handles missing values automatically.



KPI ComputationComputes dynamic profit margin fields.



High PerformanceAggregates regional and category figures in under 100ms.

> Processed `{len(df)}` Records | Total Sales: `${total_sales:,.2f}`

TECHNOLOGY ECOSYSTEM



Python

Core data analytics language used for scripting ETL workflows.



Pandas & NumPy

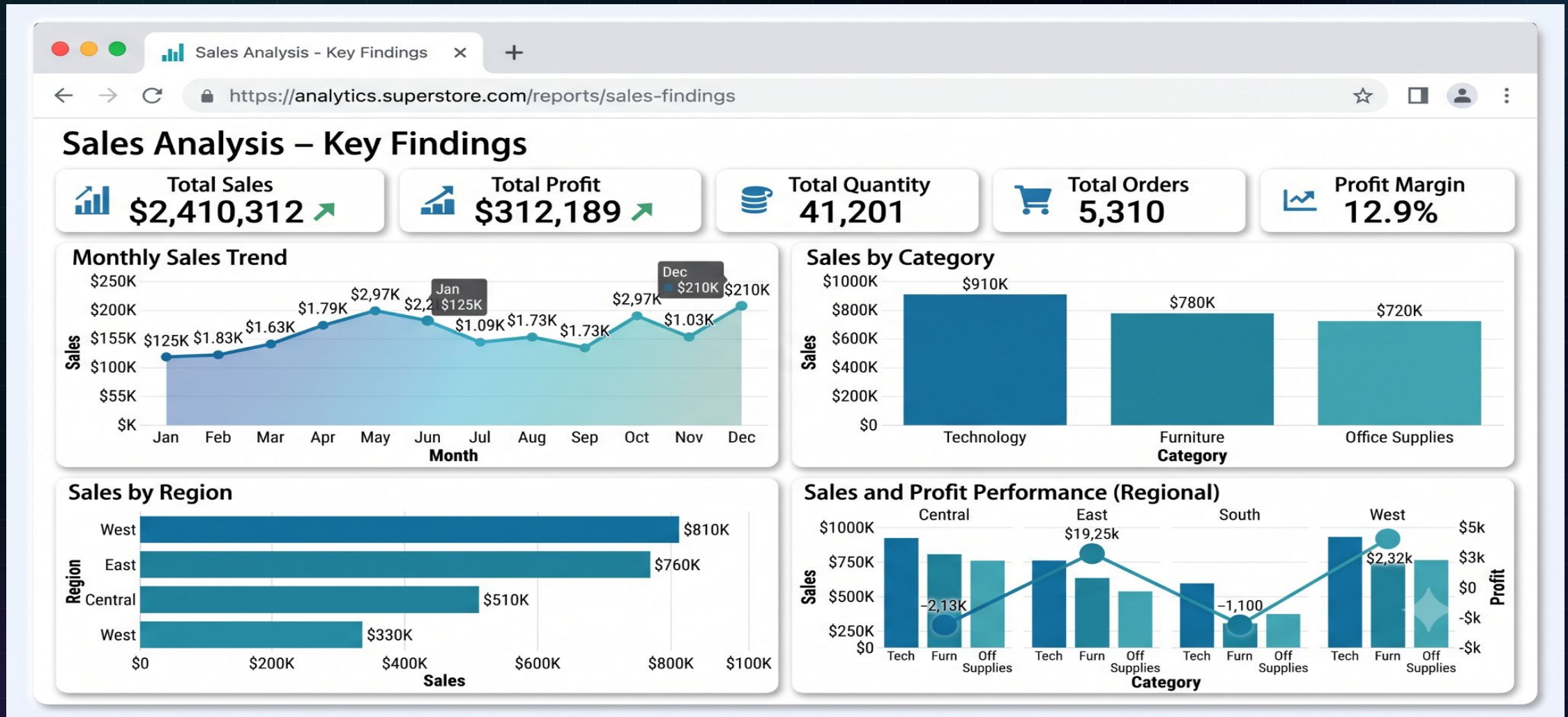
Fast data manipulation, aggregation, and mathematical computations.



Seaborn / Dashboard

Rich UI charting libraries for interactive business intelligence.

OUTPUT:



RESULTS & INSIGHTS

Key Business Discoveries

- Identified highest profit margin categories.
- Detected seasonal sales surges during Q4.
- Uncovered underperforming regional hubs.

